

Overview + Database

Hone rah kun ballashet bl project taba3 internship.

System Overview:

BabiVoyage is a RESTful API backend for a flight booking platform targeting West African air travel. The system allows users to **search** and **book** flights, **manage** their reservations, and **receive** booking confirmations — all through a secure, token-based API.

Core Features:

1- Authentication & Users:

- User registration with secure password hashing
- Token-based login/logout via Laravel Sanctum
- User profile management

2- Flights:

- Admin can create/update/delete flights
- Users can browse and search available flights
- Flights have origin, destination, departure/arrival times, price, and available seats

3- Bookings:

- Users can book available flights
- Each booking holds passenger details and seat count
- Booking status management (pending → confirmed → cancelled)

4- Passengers:

- Each booking can include multiple passengers
- Linked to the booking, not the user account directly

Database Design:

Users Table built-in in Laravel

Airports Table:

Column	Type	Notes
id	bigint PK	auto-increment
name	string	e.g. "Murtala Muhammed International"
code	string	unique, IATA code e.g. "LOS"
city	string	e.g. "Lagos"
country	string	e.g. "Nigeria"
timestamps	—	created_at, updated_at

Airlines Table:

Column	Type	Notes
id	bigint PK	auto-increment
name	string	e.g. "Air Côte d'Ivoire"
code	string	unique, IATA code e.g. "HF"
logo_url	string	nullable
timestamps	—	created_at, updated_at

Flights Table:

Column	Type	Notes
id	bigint PK	auto-increment
flight_number	string	unique, e.g. "HF204"
airline_id	FK → airlines	required
origin_airport_id	FK → airports	required
destination_airport_id	FK → airports	required
departure_time	datetime	required
arrival_time	datetime	required
price	decimal(10,2)	required
total_seats	integer	required
available_seats	integer	required
status	enum	'scheduled', 'cancelled', 'completed'
timestamps	— — —	created_at, updated_at

Bookings Table:

Column	Type	Notes
id	bigint PK	auto-increment
user_id	FK → users	required
flight_id	FK → flights	required
booking_reference	string	unique, e.g. "BV-20260330-XXXX"
seat_count	integer	how many seats booked
total_price	decimal(10,2)	seat_count × flight price
status	enum	'pending', 'confirmed', 'cancelled'
timestamps	—	created_at, updated_at

Passengers Table:

Column	Type	Notes
id	bigint PK	auto-increment
booking_id	FK → bookings	required
first_name	string	required
last_name	string	required
passport_number	string	nullable
nationality	string	nullable
date_of_birth	date	nullable
timestamps	—	created_at, updated_at

Saved_flights Table:

Column	Type	Notes
id	bigint PK 	auto-increment
user_id	FK → users 	required
flight_id	FK → flights 	required
unique(user_id, flight_id)	index	composite unique key
timestamps	—	created_at, updated_at

The Relationship Map:

Users ————— Bookings
(one user → many bookings) 1 : N

Flights ————— Bookings
(one flight → many bookings) 1 : N

Airlines ————— Flights
(one airline → many flights) 1 : N

Airports ————— Flights (as origin)
(one airport → many flights) 1 : N

Airports ————— Flights (as destination)
(one airport → many flights) 1 : N

Bookings ————— Passengers
(one booking → many passengers) 1 : N

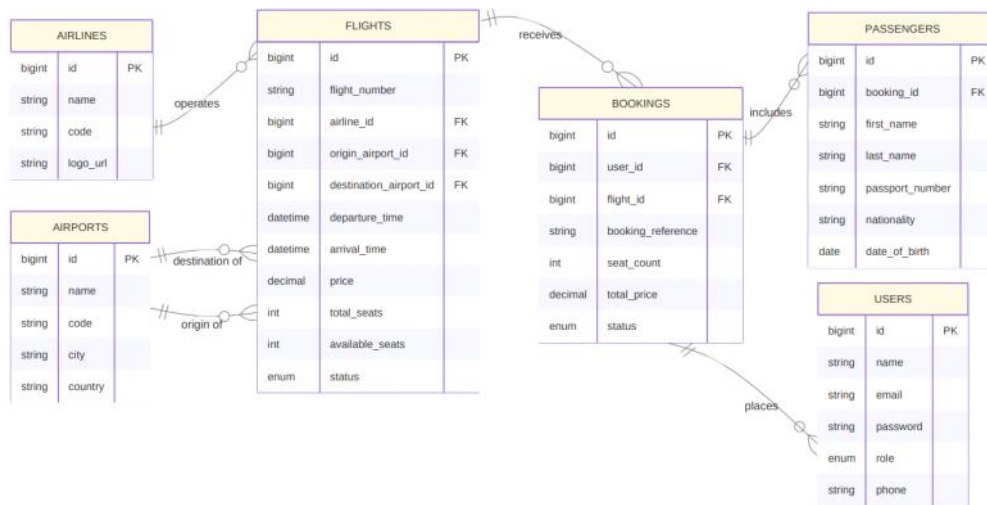
Users ————— Flights
(many users → many flights) N : N

Relationship Types Summary:

User ->Bookings	One-to-Many	bookings.user_id
Flight ->Bookings	One-to-Many	bookings.flight_id
Airline -> Flights	One-to-Many	flights.airline_id
Airport -> Flights (origin)	One-to-Many	flights.origin_airport_id
Airport -> Flights (destination)	One-to-Many	flights.destination_airport_id
Booking -> Passengers	One-to-Many	passengers.booking_id
User <-> Flights	Many-to-Many	saved_flights.user_id and saved_flights.flight_id

#	Table	Depends On
1	airlines	—
2	airports	—
3	users	—
4	flights	airlines, airports
5	bookings	users, flights
6	passengers	bookings
7	saved_flights	users, flights

Visual ERD Diagram



Repository Setup

First, I turn on the Wamp server, it should be green

Then I open the Vscode.

We can check the php and composer version to make sure all is running correctly.

In the Vscode:

- `cd C:\wamp64\www`
- `composer create-project --prefer-dist laravel/laravel webInternshipProj`

It may take some time.

```
PS C:\wamp64\www> composer create-project --prefer-dist laravel/laravel webInternshipProj

Creating migration table ..... 16.78ms DONE

[INFO] Running migrations.

0001_01_01_000000_create_users_table ..... 40.85ms DONE
0001_01_01_000001_create_cache_table ..... 23.49ms DONE
0001_01_01_000002_create_jobs_table ..... 33.61ms DONE

PS C:\wamp64\www>
```

In PHPMysqlAdmin (in wamp), we create a new database: webInternship_db

Then, in the .env file we change the name of the database

run migrations: `php artisan migrate`

It gives me this error:

SQLSTATE[42000]: Syntax error or access violation: 1071 Specified key was too long; max key length is 1000 bytes (Connection: mysql, Host: 127.0.0.1, Port: 3306, Database: webInternship_db, SQL: alter table `users` add unique `users\_email\_unique`(`email`))

How to solve it?

in the app/Providers/AppServiceProvider.php

```
public function boot(): void
{
    Schema::defaultStringLength(191);
}
```

I set the default String Length to 191, so it does not exceed the 1000 bytes limit (utf8mb4)

After doing: `php artisan migrate:fresh` => runs all migrations.

The application runs perfectly.

Repository Creation:

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj
$ git init
Initialized empty Git repository in C:/wamp64/www/webInternshipProj/.git/
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (master)
$ git add .
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (master)
$ git commit -m "Creating the Repository"
[master (root-commit) cb88df7] Creating the Repository
```

I create a new github repository "webInternshipProj" in the github website online

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (master)
$ git remote add origin https://github.com/MedianeOz/webInternshipProj.git
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (master)
$ git branch -M main
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (main)
$ git push -u origin main
Enumerating objects: 76, done.
Counting objects: 100% (76/76), done.
Delta compression using up to 4 threads
Compressing objects: 100% (59/59), done.
Writing objects: 100% (76/76), 72.42 KiB | 2.41 MiB/s, done.
Total 76 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/MedianeOz/webInternshipProj.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (main)
$ git checkout -b develop
Switched to a new branch 'develop'
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ |
```

I create a new repository online, and I link it to our project
And then I create a new branch called develop

Migrations

Now, I will create the migrations and database tables.

We already have the users table built in Laravel

This will be the creation order of the tables:

1. **airlines** (no dependencies)
2. **airports** (no dependencies)
3. **users** (already exists in Laravel)
4. **flights** (depends on: airlines, airports)
5. **bookings** (depends on: users, flights)
6. **passengers** (depends on: bookings)

Creation of Table:

```
$ php artisan make:migration create_airlines_table
php artisan make:migration create_airports_table
php artisan make:migration create_flights_table
php artisan make:migration create_bookings_table
php artisan make:migration create_passengers_table

[INFO] Migration [C:\wamp64\www\webInternshipProj\database\migrations\2026_04_24_073046_create_airlines_table.php] created successfully.
[INFO] Migration [C:\wamp64\www\webInternshipProj\database\migrations\2026_04_24_073050_create_airports_table.php] created successfully.
[INFO] Migration [C:\wamp64\www\webInternshipProj\database\migrations\2026_04_24_073050_create_flights_table.php] created successfully.
[INFO] Migration [C:\wamp64\www\webInternshipProj\database\migrations\2026_04_24_073051_create_bookings_table.php] created successfully.
[INFO] Migration [C:\wamp64\www\webInternshipProj\database\migrations\2026_04_24_073052_create_passengers_table.php] created successfully.
```

Now I change the up function of each file generated based on the tables we've created earlier.

I will only put 2 pictures

Airlines Table:

```

public function up(): void
{
    Schema::create('airlines', function (Blueprint $table) {
        $table->id();
        $table->string('name');
        $table->string('code', 10)->unique(); // IATA code e.g. "HF"
        $table->string('logo_url')->nullable();
        $table->timestamps();
    });
}

/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::dropIfExists('airlines');
}

```

Flights Table:

```

public function up(): void
{
    Schema::create('flights', function (Blueprint $table) {
        $table->id();
        $table->string('flight_number')->unique(); // e.g. "BV-101"

        $table->foreignId('airline_id')
            ->constrained('airlines')
            ->onDelete('cascade');

        $table->foreignId('origin_airport_id')
            ->constrained('airports')
            ->onDelete('restrict');

        $table->foreignId('destination_airport_id')
            ->constrained('airports')
            ->onDelete('restrict');

        $table->dateTime('departure_time');
        $table->dateTime('arrival_time');
        $table->decimal('price', 10, 2);
        $table->unsignedInteger('total_seats'); //seats can never be negative
        $table->unsignedInteger('available_seats');
        $table->enum('status', ['scheduled', 'delayed', 'cancelled', 'completed'])
            ->default('scheduled');
        $table->timestamps();
    });
}

```

I want to add 2 fields to the users table:

```

Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:migration add_fields_to_users_table
Schema: table:users

INFO Migration [C:\wamp64\www\webInternshipProj\database\migrations\2026_04_24_074803_add_fields_to_users_table.php] created successfully.

```

I create a new migration and I link it to the users table like this:

```

public function up(): void
{
    Schema::table('users', function (Blueprint $table) {
        $table->string('phone')->nullable()->after('email');
        $table->enum('role', ['user', 'admin'])->default('user')->after('phone');
    });
}

/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::table('users', function (Blueprint $table) {
        $table->dropColumn(['phone', 'role']);
    });
}

```

I want to add 2 fields to the existing users table:

```

Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:migration add_fields_to_users_table

```

Then in the up and down function:

```

public function up(): void
{
    Schema::table('users', function (Blueprint $table) {
        $table->string('phone')->nullable()->after('email');
        $table->enum('role', ['user', 'admin'])->default('user')->after('phone');
    });
}

/**
 * Reverse the migrations.
 */
public function down(): void
{
    Schema::table('users', function (Blueprint $table) {
        $table->dropColumn(['phone', 'role']);
    });
}

```

Since I only have one-to-many relationships and not many-to-many : I don't need pivot tables

But I will add a relationship between users and flights, just to add the pivot table.
A user can save many flights, a flight can be saved by many users. (many-to-many)

```

Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:migration create_saved_flights_table
INFO: Migration [C:\wamp64\www\webInternshipProj\database\migrations\2026_04_24_080948_create_saved_flights_table.php] created successfully.

```

Then:

```

public function up(): void
{
    Schema::create('saved_flights', function (Blueprint $table) {
        $table->id();

        $table->foreignId('user_id')
            ->constrained('users')
            ->onDelete('cascade');

        $table->foreignId('flight_id')
            ->constrained('flights')
            ->onDelete('cascade');

        // Prevent a user from saving the same flight twice
        $table->unique(['user_id', 'flight_id']);

        $table->timestamps();
    });
}

```

Since I added a new table, I need to adjust the ERD diagram and the previous tables

In the end, I do the migration:

```
Administrator@Mediane MINGW64 /c:/wamp64/www/webInternshipProj (develop)
$ php artisan migrate

INFO: Running migrations.

2026_04_24_073046_create_airlines_table ..... 204.28ms DONE
2026_04_24_073050_create_airports_table ..... 45.10ms DONE
2026_04_24_073050_create_flights_table ..... 226.70ms DONE
2026_04_24_073051_create_bookings_table ..... 100.66ms DONE
2026_04_24_073052_create_passengers_table ..... 41.15ms DONE
2026_04_24_074803_add_fields_to_users_table ..... 65.72ms DONE
2026_04_24_080948_create_saved_flights_table ..... 105.54ms DONE
```

The migrations done successfully

Models

Now I will create the Eloquent models

First of all, the model is singular, not like the table.

Second the User model is built-in in laravel

Now I will create 5 models: flight, Booking, Passenger, Airline and Airport

```
Administrator@Mediane MINGW64 /c:/wamp64/www/webInternshipProj (main)
$ php artisan make:model Flight
php artisan make:model Booking
php artisan make:model Passenger

[INFO] Model [C:\wamp64\www\webInternshipProj\app\Models\Flight.php] created successfully.

[INFO] Model [C:\wamp64\www\webInternshipProj\app\Models\Booking.php] created successfully.

[INFO] Model [C:\wamp64\www\webInternshipProj\app\Models\Passenger.php] created successfully.

Administrator@Mediane MINGW64 /c:/wamp64/www/webInternshipProj (develop)
$ php artisan make:model Airline

[INFO] Model [C:\wamp64\www\webInternshipProj\app\Models\Airline.php] created successfully.

Administrator@Mediane MINGW64 /c:/wamp64/www/webInternshipProj (develop)
$ php artisan make:model Airport

[INFO] Model [C:\wamp64\www\webInternshipProj\app\Models\Airport.php] created successfully.
```

Created successfully

In the User model:

I need to adjust the fillable, add use HasMany, and add the relations

```
use Database\Factories\UserFactory;
use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Foundation\Auth\User as Authenticatable;
use Illuminate\Notifications\Notifiable;
use Laravel\Sanctum\HasApiTokens;
use Illuminate\Database\Eloquent\Relations\HasMany;
use Illuminate\Database\Eloquent\Relations\BelongsToMany;

protected $fillable = [
    'name',
    'email',
    'password',
    'phone',
    'role',
];

// One user can have many bookings
public function bookings(): HasMany
{
    return $this->hasMany(Booking::class);
}

// Many-to-many: saved flights (pivot table: saved_flights)
public function savedFlights(): BelongsToMany
{
    return $this->belongsToMany(Flight::class, 'saved_flights')
        ->withTimestamps();
}
```

Now the Flight model:

```
class Flight extends Model
{
    protected $fillable = [
        'flight_number',
        'airline_id',
        'origin_airport_id',
        'destination_airport_id',
        'departure_time',
        'arrival_time',
        'price',
        'total_seats',
        'available_seats',
        'status',
    ];

    protected function casts(): array
    {
        return [
            'departure_time' => 'datetime',
            'arrival_time' => 'datetime',
            'price' => 'decimal:2',
            'total_seats' => 'integer',
            'available_seats' => 'integer',
        ];
    }
}
```

```
// One flight can have many bookings
public function bookings(): HasMany
{
    return $this->hasMany(Booking::class);
}

public function airline(): BelongsTo
{
    return $this->belongsTo(Airline::class);
}

public function originAirport(): BelongsTo
{
    return $this->belongsTo(Airport::class, 'origin_airport_id');
}

public function destinationAirport(): BelongsTo
{
    return $this->belongsTo(Airport::class, 'destination_airport_id');
}

public function savedByUsers(): BelongsToMany
{
    return $this->belongsToMany(User::class, 'saved_flights')
        ->withTimestamps();
}
```

I put all fillables from the table I created earlier

"casts()" to convert string (departure_at and arrival_at) to a carbon datetime objects.

For Booking model:

```
// A booking belongs to one user
public function user(): BelongsTo
{
    return $this->belongsTo(User::class);
}

// A booking belongs to one flight
public function flight(): BelongsTo
{
    return $this->belongsTo(Flight::class);
}

// A booking can have many passengers
public function passengers(): HasMany
{
    return $this->hasMany(Passenger::class);
}
```

For Passenger model:

```

class Passenger extends Model
{
    protected $fillable = [
        'booking_id',
        'first_name',
        'last_name',
        'passport_number',
        'nationality',
        'date_of_birth',
    ];

    protected function casts(): array
    {
        return [
            'date_of_birth' => 'date',
        ];
    }

    // A passenger belongs to one booking
    public function booking(): BelongsTo
    {
        return $this->belongsTo(Booking::class);
    }
}

```

Airline model:

```

class Airline extends Model
{
    protected $fillable = [
        'name',
        'code',
        'logo_url'
    ];

    public function flights(): HasMany
    {
        return $this->hasMany(Flight::class);
    }
}

```

Airport model:

```

class Airport extends Model
{
    protected $fillable = [
        'name',
        'code',
        'city',
        'country'
    ];

    // Flights departing from this airport
    public function originatingFlights(): HasMany
    {
        return $this->hasMany(Flight::class, 'origin_airport_id');
    }

    // Flights arriving at this airport
    public function arrivingFlights(): HasMany
    {
        return $this->hasMany(Flight::class, 'destination_airport_id');
    }
}

```

A small note:

In both Flight and User model:

we've put a function saved_flights that belongsToMany (many-to-many), that refers to the other table, as shown in the pictures.

```
User
├─ hasMany ──────────> Bookings
│
│   └─ belongsTo ─> User
│   └─ belongsTo ─> Flight
│   └─ hasMany ─> Passengers
│       └─ belongsTo ─> Booking
└─ belongsToMany ───> Flights (via saved_flights)
```

```
Flight
├─ belongsTo ─────────> Airline
├─ belongsTo ─────────> Airport (origin)
├─ belongsTo ─────────> Airport (destination)
├─ hasMany ───────────> Bookings
└─ belongsToMany ───> Users (via saved_flights)
```

```
Airline
└─ hasMany ───────────> Flights
```

```
Airport
├─ hasMany ───────────> Flights (as origin)
└─ hasMany ───────────> Flights (as destination)
```

Registration API

Here's what we need to build:

1. Create AuthController
2. Create a RegisterRequest (Form Request for validation)
3. Write the register function
4. Add the route in api.php
5. Test in Postman

AuthController:

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:controller API/AuthController -- Create the AuthController

INFO Controller [C:\wamp64\www\webInternshipProj\app\Http\Controllers\API\AuthController.php] created successfully.
```

Claude suggestion:

Instead of writing validation directly inside the controller, we use a **Form Request** — this is the professional Laravel way. It keeps the controller clean.

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:request Auth/RegisterRequest -- Create the RegisterRequest

INFO Request [C:\wamp64\www\webInternshipProj\app\Http\Requests\Auth\RegisterRequest.php] created successfully.
```

This is for the validation rules, and error messages

```
// true = everyone (since registration is public)
public function authorize(): bool
{
    return true;
}

public function rules(): array
{
    return [
        'name' => ['required', 'string', 'max:255'],
        'email' => ['required', 'string', 'email', 'max:255', 'unique:users,email'],
        'password' => ['required', 'string', 'min:8', 'confirmed'],
        'phone' => ['nullable', 'string', 'max:20'],
    ];
}

// Custom error messages (optional but professional)
public function messages(): array
{
    return [
        'name.required' => 'Full name is required.',
        'email.required' => 'Email address is required.',
        'email.email' => 'Please provide a valid email address.',
        'email.unique' => 'This email is already registered.',
        'password.required' => 'Password is required.',
        'password.min' => 'Password must be at least 8 characters.',
        'password.confirmed' => 'Password confirmation does not match.',
    ];
}
```

Why confirmed on password? Laravel's confirmed rule automatically looks for a password_confirmation field in the request and checks they match. So in Postman I'll send both password and password_confirmation.

Then I adjust the register function in AuthController

```
// by RegisterRequest - no need to validate manually here

// Create the user
$user = User::create([
    'name' => $request->name,
    'email' => $request->email,
    'password' => $request->password, // hashed automatically via cast in User model
    'phone' => $request->phone,
    'role' => 'user',
]);

// Generate Sanctum token for immediate login after registration
$token = $user->createToken('auth_token')->plainTextToken;

return response()->json([
    'success' => true,
    'message' => 'Registration successful.',
    'data' => [
        'user' => [
            'id' => $user->id,
            'name' => $user->name,
            'email' => $user->email,
            'phone' => $user->phone,
            'role' => $user->role,
        ],
        'token' => $token,
    ],
], 201);
```

In routes/api.php :

```
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\API\AuthController;

// Public routes (no token needed)
Route::post('/register', [AuthController::class, 'register']);
```

We add the route of the registration

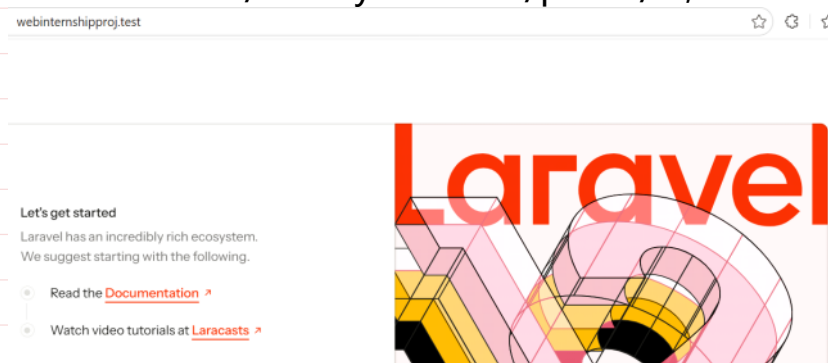
Then I test it in postman, to make sure all is running.

First I create a new collection named WebInternshipProj

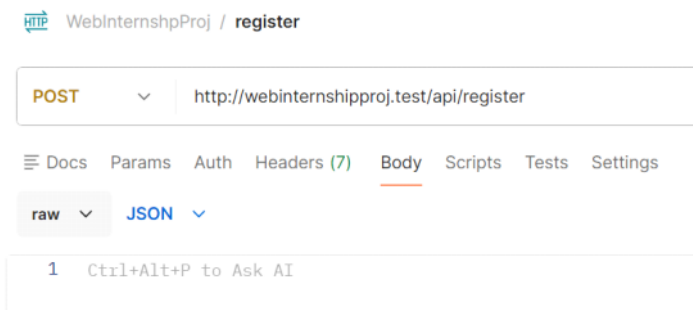
"Add a request" to create new route.

I choose post request, and name it register

Then I need to change the url of project,
from localhost/udemyWebInter/public/ , to webinternshipproj.test

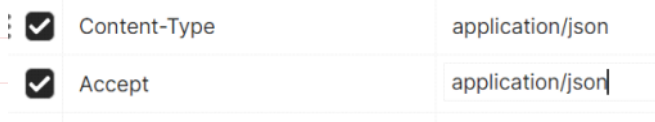


Now I put `http://webinternshipproj.test/api/register` in the url of the new post request



I check that:

Header: Content-Type is application/json | ~ | Header: Accept is application/json



Now I will be adding the data as json, and test it in postman

When I run this json data:

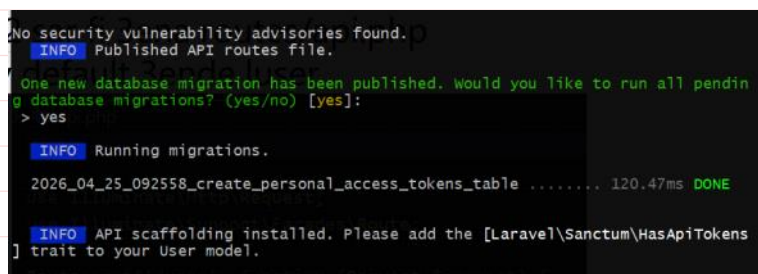
```
{
  "name": "Mediane Ozeir",
  "email": "mediane@gmail.com",
  "password": "password123",
  "password_confirmation": "password123",
  "phone": "12345678"
}
```

It gives me error, why? It tells me that there is no api/register.

So I deleted the api file I created manually,

and I used this command: **php artisan install:api**

This will create the api file in the routes, and install all dependencies needed.



It tells me to add `Laravel\Sanctum\HasApiToken` to the User model (already did)

Now this is the routes/api.php (user route created automatically)

```
<?php

use Illuminate\Http\Request;
use Illuminate\Support\Facades\Route;
use App\Http\Controllers\API\AuthController;

Route::get('/user', function (Request $request) {
    return $request->user();
})->middleware('auth:sanctum');

// Public routes (no token needed)
Route::post('/register', [AuthController::class, 'register']);
```

Now when I rerun the json code:

```
"success": true,
"message": "Registration successful.",
"data": {
  "user": {
    "id": 1,
    "name": "Mediane Ozeir",
    "email": "mediane@gmail.com",
    "phone": "12345678",
    "role": "user"
  },
  "token": "1|2fJkAZ7lC1XMg26NEUeEY4jSoVrPEV6dKg1tPHUa6902c8b"
}
```

I will send bad data and test it
no name:

```
"email": "mediane0@gmail.com",
"password": "password123",
"password_confirmation": "password123",
"phone": "12345678"
```

422 Unprocessable Content

Preview Debug with AI

```
"message": "Full name is required.",
"errors": {
  "name": [
    "Full name is required."
  ]
}
```

wrong email format:

```
"email": "medianegmail.com",
"password": "password123",
"password_confirmation": "password123",
"phone": "12345678"
```

422 Unprocessable Content · 91 ms · 45

Preview Debug with AI

```
"message": "Please provide a valid email address.",
"errors": {
  "email": [
    "Please provide a valid email address."
  ]
}
```

existing password:

```
"name": "Mediane Ozeir",  
"email": "mediane@gmail.com",  
"password": "password123",  
"password_confirmation": "password123",  
"phone": "12345678"
```

🕒 422 Unprocessable Content · 115 ms

▼ ▶ Preview 🐞 Debug with AI ▼

```
"message": "This email is already registered.",  
"errors": {  
  "email": [  
    "This email is already registered."  
  ]  
}
```

passwords unmatched:

```
"name": "Mediane Ozeir",  
"email": "medianeOz@gmail.com",  
"password": "password123",  
"password_confirmation": "password",  
"phone": "12345678"
```

🕒 422 Unprocessable Content · 104 ms · 46

▼ ▶ Preview 🐞 Debug with AI ▼

```
"message": "Password confirmation does not match.",  
"errors": {  
  "password": [  
    "Password confirmation does not match."  
  ]  
}
```

Authentication API

Here's what we need to build:

1. Create LoginRequest (and Form Request for validation)
2. Write the login function in AuthController
3. Write the logout function in AuthController
4. Add the routes in api.php
5. Test in Postman

I will start with the Login Form Request

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:request Auth/LoginRequest

INFO Request [C:\wamp64\www\webInternshipProj\app\Http\Requests\Auth>LoginRequest.php] created successfully.
Create the Login Form Request
```

In LoginRequest.php

```
public function rules(): array
{
    return [
        'email' => ['required', 'string', 'email'],
        'password' => ['required', 'string'],
    ];
}

public function messages(): array
{
    return [
        'email.required' => 'Email address is required.',
        'email.email' => 'Please provide a valid email address.',
        'password.required' => 'Password is required.',
    ];
}
```

Now I create the logic of login and logout in AuthController

Login Function:

```

public function login(LoginRequest $request)
{
    // Check if credentials are correct
    if (!Auth::attempt($request->only('email', 'password'))) {
        return response()->json([
            'success' => false,
            'message' => 'Invalid credentials. Please check your email and password.',
        ], 401);
    }

    // Get the authenticated user
    $user = Auth::user();

    // Revoke all old tokens (optional but clean - one session at a time)
    $user->tokens()->delete();

    // Generate a fresh token
    $token = $user->createToken('auth_token')->plainTextToken;

    return response()->json([
        'success' => true,
        'message' => 'Login successful.',
        'data' => [
            'user' => [
                'id' => $user->id,
                'name' => $user->name,
                'email' => $user->email,
                'phone' => $user->phone,
                'role' => $user->role,
            ],
            'token' => $token,
        ],
    ], 200);
}

```

Logout Function

```

public function logout()
{
    // Revoke all tokens for the currently authenticated user
    Auth::user()->tokens()->delete();

    return response()->json([
        'success' => true,
        'message' => 'Logged out successfully.',
    ], 200);
}

```

In routes/api:

```

// -----
// Public routes (no token needed)
// -----

Route::post('/register', [AuthController::class, 'register']);
Route::post('/login', [AuthController::class, 'login']);

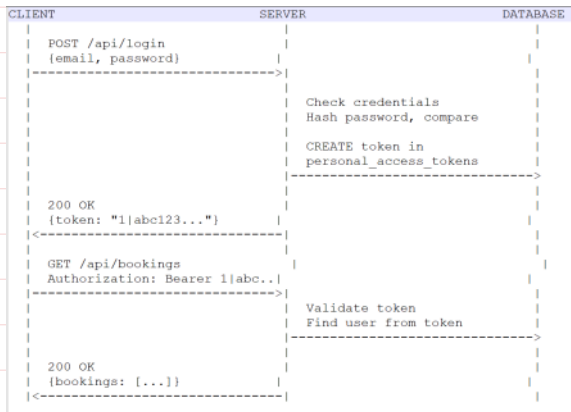
// -----
// Protected Routes (token required)
// -----

Route::middleware('auth:sanctum')->group(function () {
    Route::post('/logout', [AuthController::class, 'logout']);
});

```

In the login, the token is created, and then it is used to do any action, and then it is deleted in the logout (to do the logout we need the token)

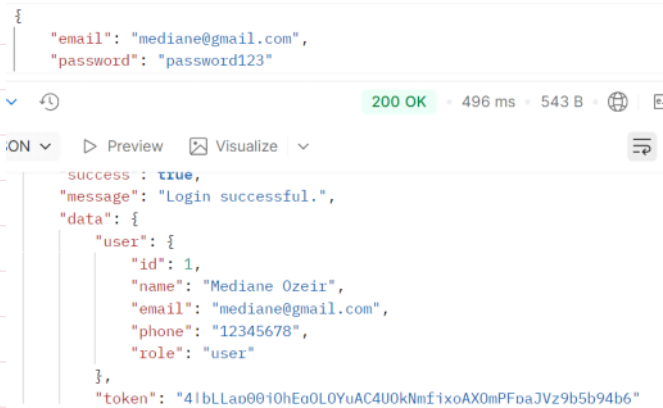
Why middleware('auth:sanctum') on logout? Because I need to be logged in to log out. If a user sends a request to /logout without a valid token, Sanctum automatically returns 401 Unauthenticated — I don't need to handle that manually.



Now I will test it in Postman:

I create a new post request with url /api/login

Login Successful



Wrong Password:

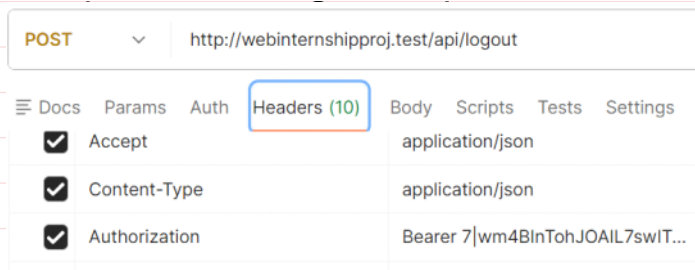


Now for Logout:

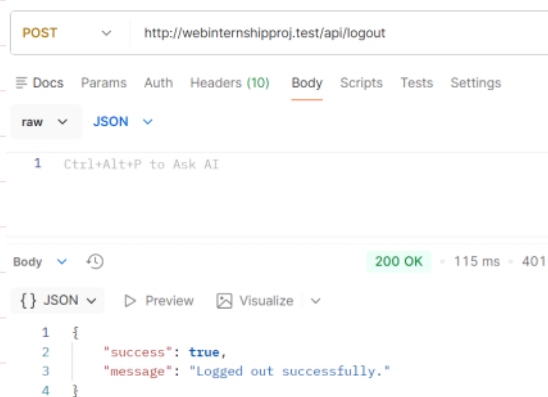
I create a new post logout request with url /api/logout

But here I need to add one thing to the header

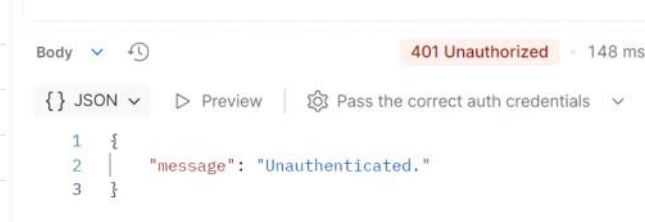
When I login, it creates a specific token, I will copy it
And paste it the logout request header, like this:



So when I send the logout request, I don't send anything in the body, it does it based on the token of the login.



If I remove the token from the header, it will give me this error:



How the token flow works:

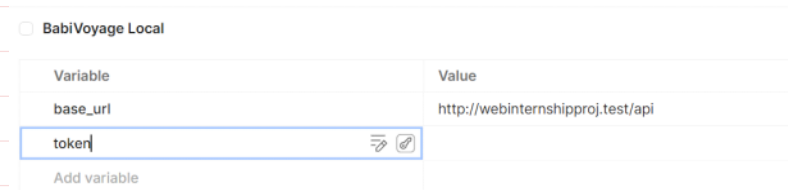
```
POST /register → creates user → returns token
POST /login    → checks credentials → deletes old tokens → returns new token
POST /logout   → requires token in header → deletes all tokens → logged out
```

Postman Collection

Now I'll adjust the postman collection, and create all endpoints needed

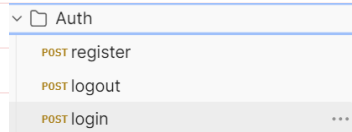
I will use environment variables (base_url, token).

I create a postman environment: it stores variables that change based on where I run the API

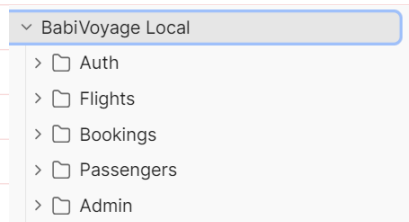


Like this, I don't need to every time right the whole url, I can just write base_url
And for the token, it will be saved and deleted automatically (based on a code I will write it later on).

I create a folder Auth, and drag the three requests: login, logout and register into it.



I create the folders:





Like this, when the user register, the token is stored automatically in the token variable created in the env.

It should be done in the Post-response and not Pre-request (when I write it first in the pre-request it gave me error).

The same for login:



For the logout:



It will automatically get the token from the Beared `{{token}}`, and delete it.

Now I will test it:

Register:

```

1 {
2   "name": "Jad Ozeir",
3   "email": "jadozeir@gmail.com",
4   "password": "password123",
5   "password_confirmation": "password123",
6   "phone": "12345678"

```

/ Cookies Headers 9 Test Results 1/1

JSON Preview Visualize

```

1 {
2   "success": true,
3   "message": "Registration successful.",
4   "data": {
5     "user": {
6       "id": 4,
7       "name": "Jad Ozeir",
8       "email": "jadozeir@gmail.com",
9       "phone": "12345678",
10      "role": "user"
11    }
12  },

```

Login:

```

1 {
2   "email": "jadozeir@gmail.com",
3   "password": "password123"

```

y Cookies Headers 9 Test Results 1/1

200 OK · 434 ms

JSON Preview Visualize

```

1 {
2   "success": true,
3   "message": "Login successful.",
4   "data": {
5     "user": {
6       "id": 4,
7       "name": "Jad Ozeir",
8       "email": "jadozeir@gmail.com",
9       "phone": "12345678",
10      "role": "user"
11    },
12   "token": "10|VqTJyzHBN3k5A3PDck91NtgXBWp0AFkS5aca7XJu06dfd46b"
13 }

```

Logout:

Body Cookies Headers 10 Test Results 1/1

{ } JSON Preview Visualize

```

1 {
2   "success": true,
3   "message": "Logged out successfully."
4 }

```

I created 2 requests in Auth folder, getProfile and changeProfile, and it takes token in the variables.

I created 3 requests in the Flights folder, that doesn't need token.

```

POST getProfile
PUT changeProfile
Flights
GET listFlights
GET getOneFlight
GET searchFlights

```

These 3 requests are done only by the admin, and requires token:

POST addFlights

PUT updateFlight

DEL deleteFlight

Since all requests in the Bookings and Passengers folder requires token, I will add it into the authorization of the whole folder like this:

BabiVoyage Local > Bookings

Overview Authorization Scripts

This authorization method will be used for every request in this folder. You can override this by specifying one in the request.

Auth Type

Bearer Token

The authorization header will be automatically generated when you send the request.

Learn more about [authorization](#)

Token

{{token}}

Bookings

GET myBookings

POST createBooking

GET getOneBooking

PATCH cancelBooking

Passengers

GET listPassengers

POST addPassengers

DEL deletePassenger

I will add a sample of the json code:

Create Flights (Admin):

POST {{base_url}}/admin/flights

Docs Params Authorization Headers 11 Body Scr

none form-data x-www-form-urlencoded raw

```
1 {
2   "flight_number": "HF204",
3   "airline_id": 1,
4   "origin_airport_id": 1,
5   "destination_airport_id": 2,
6   "departure_time": "2026-05-01 08:00:00",
7   "arrival_time": "2026-05-01 10:30:00",
8   "price": 150.00,
9   "total_seats": 120,
10  "available_seats": 120,
11  "status": "scheduled"
12 }
```

Create Booking:

POST ▼ {{base_url}} /bookings

☰ Docs Params Authorization Headers 1

☐ none ☐ form-data ☐ x-www-form-urlencoded

```
1 {  
2   "flight_id": 1,  
3   "seat_count": 2  
4 }
```

Add Passenger:

POST ▼ {{base_url}} /bookings/{id}/passengers

☰ Docs Params Authorization Headers 11 Body ●

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒

```
1 {  
2   "first_name": "Mediane",  
3   "last_name": "Ozeir",  
4   "passport_number": "A123456",  
5   "nationality": "Lebanese",  
6   "date_of_birth": "2002-05-15"  
7 }
```



WebInter
nshpPro...

Full CRUD API

- Phase 1: Flight CRUD (FlightController + Form Requests + Routes)
- Phase 2: Booking CRUD (BookingController + Form Requests + Routes)
- Phase 3: Passenger CRUD (PassengerController + Form Requests + Routes)
- Phase 4: Update Postman (Test everything)

Phase 1: Flight CRUD

First, I will create the FlightController:

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:controller API/FlightController

INFO Controller [C:\wamp64\www\webInternshipProj\app\Http\Controllers\API\FlightController.php] created successfully.
```

Flight Form Request:

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:request Flight/StoreFlightRequest
php artisan make:request Flight/UpdateFlightRequest

INFO Request [C:\wamp64\www\webInternshipProj\app\Http\Requests\Flight\StoreFlightRequest.php] created successfully.

INFO Request [C:\wamp64\www\webInternshipProj\app\Http\Requests\Flight\UpdateFlightRequest.php] created successfully.
```

In the **StoreFlightRequest**:

Create Flights:

```
public function authorize(): bool
{
    return auth()->check() && auth()->user()->role === 'admin'; //only admins can create flights
}

public function rules(): array
{
    return [
        'flight_number' => ['required', 'string', 'max:20', 'unique:flights,flight_number'],
        'airline_id' => ['required', 'integer', 'exists:airlines,id'],
        'origin_airport_id' => ['required', 'integer', 'exists:airports,id'],
        'destination_airport_id' => ['required', 'integer', 'exists:airports,id', 'different:origin_airport_id'],
        'departure_time' => ['required', 'date', 'after:now'],
        'arrival_time' => ['required', 'date', 'after:departure_time'],
        'price' => ['required', 'numeric', 'min:0'],
        'total_seats' => ['required', 'integer', 'min:1'],
        'available_seats' => ['required', 'integer', 'min:0', 'lte:total_seats'],
        'status' => ['required', 'in:scheduled,delayed,cancelled,completed'],
    ];
}

public function messages(): array
{
    return [
        'flight_number.unique' => 'This flight number already exists.',
        'destination_airport_id.different' => 'Destination must be different from origin.',
        'departure_time.after' => 'Departure time must be in the future.',
        'arrival_time.after' => 'Arrival time must be after departure time.',
        'available_seats.lte' => 'Available seats cannot exceed total seats.',
        'status.in' => 'Status must be: scheduled, delayed, cancelled, or completed.',
    ];
}
```

UpdateFlightRequest:

```
public function authorize(): bool
{
    return auth()->check() && auth()->user()->role === 'admin'; //only admin can update flights
}

public function rules(): array
{
    $flightId = $this->route('id'); //updating specific flight

    return [
        'flight_number' => ['sometimes', 'string', 'max:20', Rule::unique('flights', 'flight_number')->ignore($flightId)],
        'airline_id' => ['sometimes', 'integer', 'exists:airlines,id'],
        'origin_airport_id' => ['sometimes', 'integer', 'exists:airports,id'],
        'destination_airport_id' => ['sometimes', 'integer', 'exists:airports,id', 'different:origin_airport_id'],
        'departure_time' => ['sometimes', 'date'],
        'arrival_time' => ['sometimes', 'date', 'after:departure_time'],
        'price' => ['sometimes', 'numeric', 'min:0'],
        'total_seats' => ['sometimes', 'integer', 'min:1'],
        'available_seats' => ['sometimes', 'integer', 'min:0'],
        'status' => ['sometimes', 'in:scheduled,delayed,cancelled,completed'],
    ];
}
```

Why sometimes? On update, the admin might only want to change the price. sometimes means: only validate this field if it's present in the request.

Now I will create the **FlightController**.

It contains: list, search, get specific, create, update and delete.

1- List Flights {index}:

```
// -----
// PUBLIC: GET /api/flights
// List all available (scheduled) flights
// -----
public function index()
{
    $flights = Flight::with(['airline', 'originAirport', 'destinationAirport'])
        ->where('status', 'scheduled')
        ->get();

    return response()->json([
        'success' => true,
        'data' => $flights,
    ], 200);
}
```

2- Search Flights {search}:

(Flights can be filtered with: Origin, Destination, Departure time, Airport code or Price)

```

public function search(Request $request)
{
    $request->validate([
        'origin' => ['nullable', 'string'],
        'destination' => ['nullable', 'string'],
        'date' => ['nullable', 'date'],
        'min_price' => ['nullable', 'numeric', 'min:0'],
        'max_price' => ['nullable', 'numeric', 'min:0'],
    ]);

    $query = Flight::with(['airline', 'originAirport', 'destinationAirport'])
        ->where('status', 'scheduled');

    // Filter by origin city or airport code
    if ($request->filled('origin')) {
        $query->whereHas('originAirport', function ($sq) use ($request) {
            $sq->where('city', 'like', '%' . $request->origin . '%')
                ->orWhere('code', 'like', '%' . $request->origin . '%');
        });
    }

    // Filter by destination city or airport code
    if ($request->filled('destination')) {
        $query->whereHas('destinationAirport', function ($sq) use ($request) {
            $sq->where('city', 'like', '%' . $request->destination . '%')
                ->orWhere('code', 'like', '%' . $request->destination . '%');
        });
    }

    // Filter by departure date
    if ($request->filled('date')) {
        $query->whereDate('departure_time', $request->date);
    }

    // Filter by price range
    if ($request->filled('min_price')) {
        $query->where('price', '>=', $request->min_price);
    }

    if ($request->filled('max_price')) {
        $query->where('price', '<=', $request->max_price);
    }

    $flights = $query->get();

    return response()->json([
        'success' => true,
        'data' => $flights,
    ], 200);
}

```

3- Get a specific flight {show}:

```

// -----
// PUBLIC: GET /api/flights/{id}
// Get a specific flight
// -----
public function show($id)
{
    $flight = Flight::with(['airline', 'originAirport', 'destinationAirport'])
        ->find($id);

    if (!$flight) {
        return response()->json([
            'success' => false,
            'message' => 'Flight not found.',
        ], 404);
    }

    return response()->json([
        'success' => true,
        'data' => $flight,
    ], 200);
}

```

4- Create a new Flight {store}:

```
// -----
// ADMIN: POST /api/admin/flights
// Create a new flight
// -----
public function store(StoreFlightRequest $request)
{
    $flight = Flight::create($request->validated());

    return response()->json([
        'success' => true,
        'message' => 'Flight created successfully.',
        'data' => $flight->load(['airline', 'originAirport', 'destinationAirport']),
    ], 201);
}
```

The work is simplified and well-organized because of the request forms.

5- Update a Flight {update}:

```
// -----
// ADMIN: PUT /api/admin/flights/{id}
// Update a flight
// -----
public function update(UpdateFlightRequest $request, $id)
{
    $flight = Flight::find($id);

    if (!$flight) {
        return response()->json([
            'success' => false,
            'message' => 'Flight not found.',
        ], 404);
    }

    $flight->update($request->validated());

    return response()->json([
        'success' => true,
        'message' => 'Flight updated successfully.',
        'data' => $flight->load(['airline', 'originAirport', 'destinationAirport']),
    ], 200);
}
```

6- Delete a Flight {destroy}:

```
// -----
// ADMIN: DELETE /api/admin/flights/{id}
// Delete a flight
// -----
public function destroy($id)
{
    $flight = Flight::find($id);

    if (!$flight) {
        return response()->json([
            'success' => false,
            'message' => 'Flight not found.',
        ], 404);
    }

    $flight->delete();

    return response()->json([
        'success' => true,
        'message' => 'Flight deleted successfully.',
    ], 200);
}
```

I've finished this part for now:

- Flights
 - GET listFlights
 - GET getOneFlight
 - GET searchFlights
 - POST addFlights
 - PUT updateFlight
 - DEL deleteFlight

Before creating the booking and passenger controllers, I want to create the routes for the FlightController functions:

```
// Public flight browsing (anyone can browse)
Route::get('/flights', [FlightController::class, 'index']);
Route::get('/flights/search', [FlightController::class, 'search']);
Route::get('/flights/{id}', [FlightController::class, 'show']);

// -----
// Protected Routes (token required)
// -----

Route::middleware('auth:sanctum')->group(function () {

    // Auth
    Route::post('/logout', [AuthController::class, 'logout']);

    // Admin: Flight management
    Route::post('/admin/flights', [FlightController::class, 'store']);
    Route::put('/admin/flights/{id}', [FlightController::class, 'update']);
    Route::delete('/admin/flights/{id}', [FlightController::class, 'destroy']);
});
```

Now I will check all functions in postman, to make sure everything is running successfully.

For the data of Airlines and Airports:
I will manually use Database Seeders.

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:seeder AirlineSeeder

INFO Seeder [C:\wamp64\www\webInternshipProj\database\seeders\AirlineSeeder.
php] created successfully.

Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:seeder AirportSeeder

INFO Seeder [C:\wamp64\www\webInternshipProj\database\seeders\AirportSeeder.
php] created successfully.
```

database/seeders/AirlineSeeder:

```
class AirlineSeeder extends Seeder
{
    public function run(): void
    {
        Airline::insert([
            ['name' => "Air Côte d'Ivoire", 'code' => 'HF', 'logo_url' => null],
            ['name' => 'Askya Airlines', 'code' => 'KP', 'logo_url' => null],
            ['name' => 'Air Senegal', 'code' => 'HC', 'logo_url' => null],
            ['name' => 'Arik Air', 'code' => 'W3', 'logo_url' => null],
            ['name' => 'Air Burkina', 'code' => '2J', 'logo_url' => null],
            ['name' => 'Ceiba Intercontinental', 'code' => 'C2', 'logo_url' => null],
            ['name' => 'Cronos Airlines', 'code' => 'C8', 'logo_url' => null],
        ]);
    }
}
```

AirportSeeder:

```
class AirportSeeder extends Seeder
{
    public function run(): void
    {
        Airport::insert([
            [
                'name' => "Félix Houphouët-Boigny International Airport",
                'code' => 'ABJ',
                'city' => 'Abidjan',
                'country' => "Côte d'Ivoire",
            ],
            [
                'name' => 'Kotoka International Airport',
                'code' => 'ACC',
                'city' => 'Accra',
                'country' => 'Ghana',
            ],
            [
                'name' => 'Murtala Muhammed International Airport',
                'code' => 'LOS',
                'city' => 'Lagos',
                'country' => 'Nigeria',
            ],
        ], false);
    }
}
```

DatabaseSeeder:

```
class DatabaseSeeder extends Seeder
{
    use WithoutModelEvents;

    /**
     * Seed the application's database.
     */
    public function run(): void
    {
        $this->call([
            AirlineSeeder::class,
            AirportSeeder::class,
        ]);
    }
}
```

Database created successfully

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan db:seed

INFO Seeding database.
Database\Seeders\AirlineSeeder ..... RUNNING
Database\Seeders\AirlineSeeder ..... 65 ms DONE
Database\Seeders\AirportSeeder ..... RUNNING
Database\Seeders\AirportSeeder ..... 5 ms DONE
```

Now let's continue

TEST in Postman:

- Admin create a new flight:

```
{
  "flight_number": "HF204",
  "airline_id": 1,
  "origin_airport_id": 1,
  "destination_airport_id": 2,
  "departure_time": "2026-05-01 08:00:00",
  "arrival_time": "2026-05-01 10:30:00",
  "price": 150.00,
  "total_seats": 120,
  "available_seats": 120,
  "status": "scheduled"
}
```

It gives me:

```
Body Cookies Headers 10 Test Results ⌵
[{} JSON ▾ ▶ Preview ⚙ Pass the correct auth credentials
1 {
2   "message": "Unauthenticated."
3 }
```

Unauthenticated because I tried to create a new flight as a user not admin.
I changed the role manually from user to admin and logged-in again.

Now:

```
"success": true,
"message": "Login successful.",
"data": {
  "user": {
    "id": 1,
    "name": "Mediane Ozeir",
    "email": "mediane@gmail.com",
    "phone": "12345678",
    "role": "admin"
```

```
"success": true,
"message": "Flight created successfully.",
"data": {
  "flight_number": "HF204",
  "airline_id": 1,
  "origin_airport_id": 1,
  "destination_airport_id": 2,
  "departure_time": "2026-05-01T08:00:00.000000Z",
  "arrival_time": "2026-05-01T10:30:00.000000Z",
  "price": "150.00",
  "total_seats": 120,
  "available_seats": 120,
  "status": "scheduled",
  "updated_at": "2026-04-29T10:54:02.000000Z",
  "created_at": "2026-04-29T10:54:02.000000Z",
  "id": 2,
  "airline": {
    "id": 1,
    "name": "Air Côte d'Ivoire",
    "code": "HF",
    "logo_url": null,
    "created_at": null,
    "updated_at": null
  },
  "origin_airport": {
    "id": 1,
    "name": "Félix Houphouët-Boigny International Airport",
    "code": "ABJ",
    "city": "Abidjan",
    "country": "Côte d'Ivoire",
    "created_at": null,
    "updated_at": null
  },
  "destination_airport": {
    "id": 2,
    "name": "Kotoka International Airport",
    "code": "ACC",
    "city": "Accra",
```

Flight added successfully

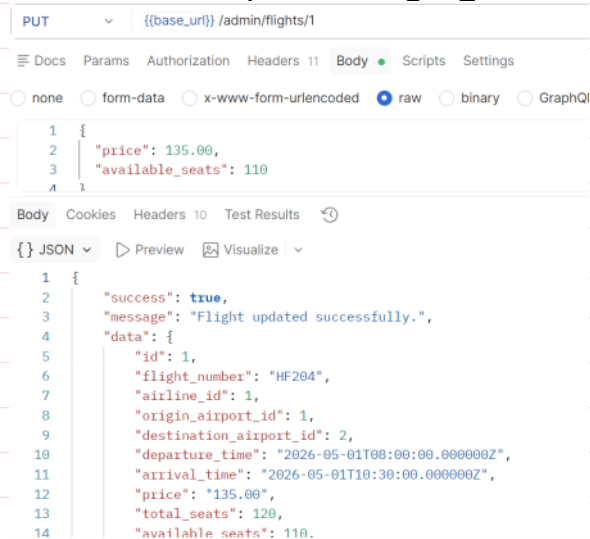
```
7   "status": "scheduled",
8   "price": 145.00,
9   "total_seats": 100,
10  "available_seats": 85,
11  "status": "nothing"
12 }
13
dy Cookies Headers 10 Test Results ⌵ 422 Unprocessable Content 1
JSON ▾ ▶ Preview ⚙ Debug with AI ▾
1 {
2   "message": "Status must be: scheduled, delayed, cancelled, or completed.",
3   "errors": {
4     "status": [
```

It gives me error, because the status is wrong, as shown in the message.

Update Flight:

(I can send only the data that I want to change, not necessary all data as mentioned before)

Also I need to put the flight_id in the url

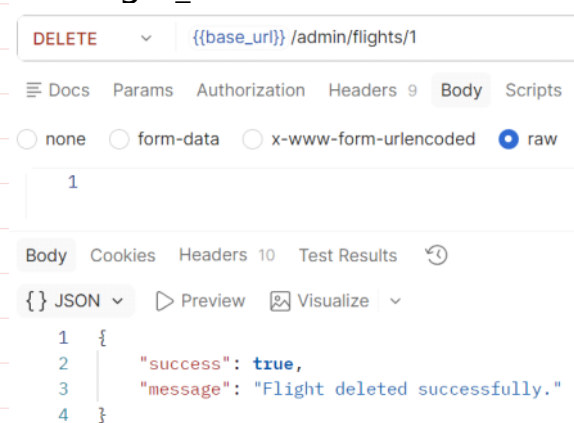


If I put the price as string it gives me error:

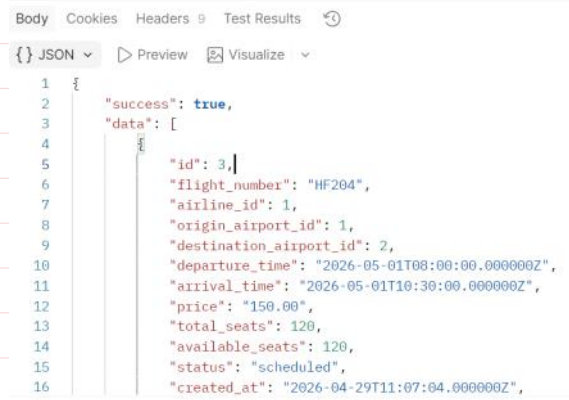


Now delete this flight:

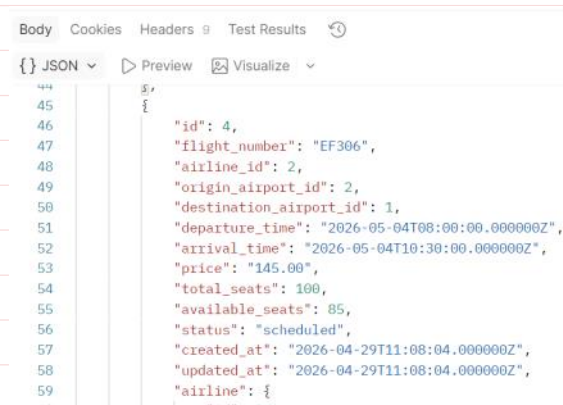
Also flight_id in the url:



List Flights: (No JSON code needed)

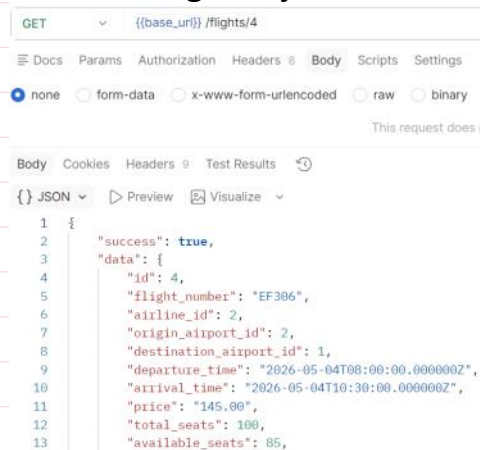


```
1 {
2   "success": true,
3   "data": [
4     {
5       "id": 3,
6       "flight_number": "HF204",
7       "airline_id": 1,
8       "origin_airport_id": 1,
9       "destination_airport_id": 2,
10      "departure_time": "2026-05-01T08:00:00.000000Z",
11      "arrival_time": "2026-05-01T10:30:00.000000Z",
12      "price": "150.00",
13      "total_seats": 120,
14      "available_seats": 120,
15      "status": "scheduled",
16      "created_at": "2026-04-29T11:07:04.000000Z",
```



```
44 {
45   "id": 4,
46   "flight_number": "EF306",
47   "airline_id": 2,
48   "origin_airport_id": 2,
49   "destination_airport_id": 1,
50   "departure_time": "2026-05-04T08:00:00.000000Z",
51   "arrival_time": "2026-05-04T10:30:00.000000Z",
52   "price": "145.00",
53   "total_seats": 100,
54   "available_seats": 85,
55   "status": "scheduled",
56   "created_at": "2026-04-29T11:08:04.000000Z",
57   "updated_at": "2026-04-29T11:08:04.000000Z",
58   "airline": {
59     "id": 2,
```

Get one Flight, by id:



```
1 {
2   "success": true,
3   "data": {
4     "id": 4,
5     "flight_number": "EF306",
6     "airline_id": 2,
7     "origin_airport_id": 2,
8     "destination_airport_id": 1,
9     "departure_time": "2026-05-04T08:00:00.000000Z",
10    "arrival_time": "2026-05-04T10:30:00.000000Z",
11    "price": "145.00",
12    "total_seats": 100,
13    "available_seats": 85,
```

Search Flights:

There is 2 methods to get data:

- 1- Put the parameters in the Param tab in postman
- 2- Put the parameters manually in the url

Method 1:

Query Params	
Key	Value
✓ origin	ACC
Key	Value

Body	Cookies	Headers	9	Test Results	🔄
------	---------	---------	---	--------------	---

{} JSON

Preview

Visualize

```

1  {
2    "success": true,
3    "data": [
4      {
5        "id": 4,
6        "flight_number": "EF306",
7        "airline_id": 2,
8        "origin_airport_id": 2,
9        "destination_airport_id": 1,
10       "departure_time": "2026-05-04T08:00:00.000000Z",
11       "arrival_time": "2026-05-04T10:30:00.000000Z",

```

Method 2:

GET	{{base_url}} /flights/search?origin=ACC
-----	---

Docs

Params

Authorization

Headers

8

Body

Scripts

Settings

☒ none
☐ form-data
☐ x-www-form-urlencoded
☐ raw
☐ binary
☐ Gr

This request does not have a body

Body	Cookies	Headers	9	Test Results	🔄
------	---------	---------	---	--------------	---

{} JSON

Preview

Visualize

```

1  {
2    "success": true,
3    "data": [
4      {
5        "id": 4,
6        "flight_number": "EF306",
7        "airline_id": 2,
8        "origin_airport_id": 2,
9        "destination_airport_id": 1,
10       "departure_time": "2026-05-04T08:00:00.000000Z",
11       "arrival_time": "2026-05-04T10:30:00.000000Z",
12       "price": "145.00",

```

Method 1 is better and more professional.

For now, all these requests are working successfully:

Auth
POST register
POST logout
POST login
POST getProfile
PUT changeProfile
Flights
GET listFlights
GET getOneFlight
GET searchFlights
POST addFlights
PUT updateFlight
DEL deleteFlight

Phase 2: Booking CRUD

First I create the BookingController and the StoreBookingRequest

```
Administrator@Mediane MINGW64 /c:/wamp64/www/webInternshipProj (develop)
$ php artisan make:controller API/BookingController

[INFO] Controller [C:\wamp64\www\webInternshipProj\app\Http\Controllers\API\Bo
okingController.php] created successfully.

Administrator@Mediane MINGW64 /c:/wamp64/www/webInternshipProj (develop)
$ php artisan make:request Booking/StoreBookingRequest

[INFO] Request [C:\wamp64\www\webInternshipProj\app\Http\Requests\Booking\Stor
eBookingRequest.php] created successfully.
```

StoreBookingRequest:

```
public function authorize(): bool
{
    return auth()->check(); //authenticated user can book a flight, not a guest
}

public function rules(): array
{
    return [
        'flight_id' => ['required', 'integer', 'exists:flights,id'],
        'seat_count' => ['required', 'integer', 'min:1', 'max:10'],
    ];
}

public function messages(): array
{
    return [
        'flight_id.exists' => 'The selected flight does not exist.',
        'seat_count.min' => 'You must book at least 1 seat.',
        'seat_count.max' => 'You cannot book more than 10 seats at once.',
    ];
}
```

BookingController:

1- Get all flights of a user {index}

```
// -----
// GET /api/bookings
// List all bookings for the authenticated user
// -----
public function index()
{
    $bookings = Booking::with(['flight.airline', 'flight.originAirport', 'flight.destinationAirport', 'passengers'])
        ->where('user_id', auth()->id())
        ->latest()
        ->get();

    return response()->json([
        'success' => true,
        'data' => $bookings,
    ], 200);
}
```

2- Create new booking {store}

```
// POST /api/bookings
// Create a new booking
// -----
public function store(StoreBookingRequest $request)
{
    $flight = Flight::find($request->flight_id);

    // Check flight is available
    if ($flight->status !== 'scheduled') {
        return response()->json([
            'success' => false,
            'message' => 'This flight is not available for booking.',
        ], 422);
    }

    // Check enough seats available
    if ($flight->available_seats < $request->seat_count) {
        return response()->json([
            'success' => false,
            'message' => 'Not enough seats available. Only ' . $flight->available_seats . ' seats left.',
        ], 422);
    }

    // Calculate total price
    $totalPrice = $flight->price * $request->seat_count;
```

```

// Create the booking
$booking = Booking::create([
    'user_id'      => auth()->id(),
    'flight_id'    => $flight->id,
    'booking_reference' => 'BV-' . strtoupper(Str::random(8)),
    'seat_count'   => $request->seat_count,
    'total_price'  => $totalPrice,
    'status'       => 'confirmed',
]);

// Reduce available seats on the flight
$flight->decrement('available_seats', $request->seat_count);

return response()->json([
    'success' => true,
    'message' => 'Booking created successfully.',
    'data'    => $booking->load(['flight.airline', 'flight.originAirport', 'flight.destinationAirport']),
], 201);
}

```

3- Get specific booking {show}: only owner of the booking can see it.

```

// Get a specific booking (only owner can see it)
// -----
public function show($id)
{
    $booking = Booking::with(['flight.airline', 'flight.originAirport', 'flight.destinationAirport', 'passengers'])
        ->find($id);

    if (!$booking) {
        return response()->json([
            'success' => false,
            'message' => 'Booking not found.',
        ], 404);
    }

    // Authorization: only the owner can view their booking
    if ($booking->user_id !== auth()->id()) {
        return response()->json([
            'success' => false,
            'message' => 'You are not authorized to view this booking.',
        ], 403);
    }

    return response()->json([
        'success' => true,
        'data'    => $booking,
    ], 200);
}

```

4- Cancel a booking

```

// PATCH /api/bookings/{id}/cancel
// Cancel a booking
// -----
public function cancel($id)
{
    $booking = Booking::find($id);

    if (!$booking) {
        return response()->json([
            'success' => false,
            'message' => 'Booking not found.',
        ], 404);
    }

    // Authorization check
    if ($booking->user_id !== auth()->id()) {
        return response()->json([
            'success' => false,
            'message' => 'You are not authorized to cancel this booking.',
        ], 403);
    }

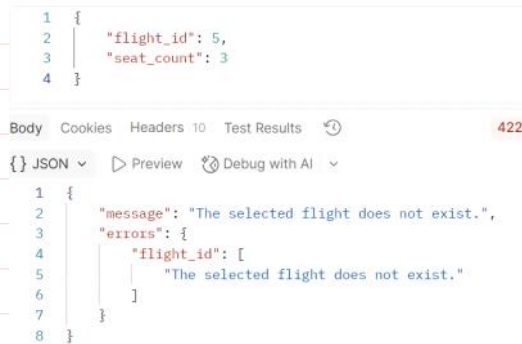
    // Check it's not already cancelled
    if ($booking->status === 'cancelled') {
        return response()->json([
            'success' => false,
            'message' => 'This booking is already cancelled.',
        ], 422);
    }
}

```

Now I will add the routes for the booking functions, and then test it in Postman

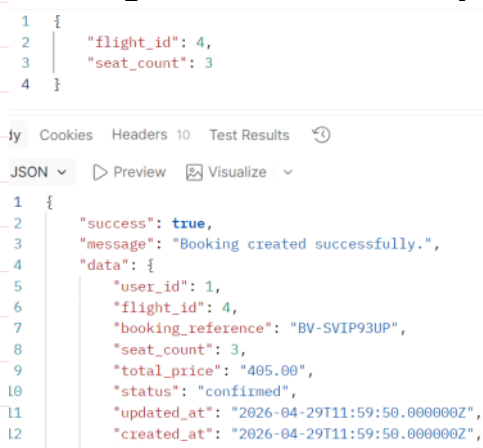
```
// Bookings
Route::get('/bookings', [BookingController::class, 'index']);
Route::post('/bookings', [BookingController::class, 'store']);
Route::get('/bookings/{id}', [BookingController::class, 'show']);
Route::patch('/bookings/{id}/cancel', [BookingController::class, 'cancel']);
```

Testing:

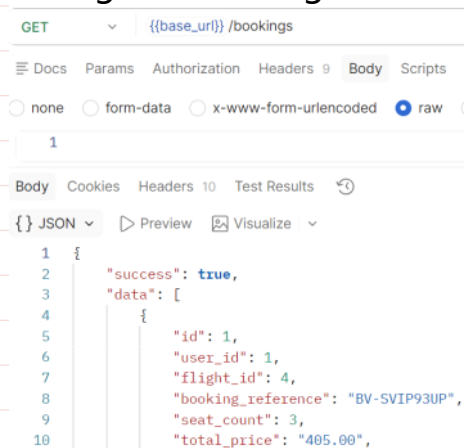


It gives me error because there is no flight with id 5.

Booking created successfully



Listing the booking of an authenticated user:



Get one booking with id:2

GET `{{base_url}}/bookings/2`

Docs Params Authorization Headers 9 Body Scripts Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary

1

Body Cookies Headers 10 Test Results

`{}` JSON `>` Preview `<img alt="visualize icon" data-bbox="238 211 251 221"/> Visualize >`

```
1 {
2   "success": true,
3   "data": {
4     "id": 2,
5     "user_id": 1,
6     "flight_id": 3,
7     "booking_reference": "BV-TUZIP8CZ3",
8     "seat_count": 3,
9     "total_price": "450.00",
10    "status": "confirmed",
11    "created_at": "2026-04-29T12:03:47.000000Z",
```

Cancel a booking: /2/cancel

PATCH `{{base_url}}/bookings/2/cancel`

Docs Params Authorization Headers 10 Body Scripts Settings

☒ none ☐ form-data ☐ x-www-form-urlencoded ☐ raw ☐ binary

This request does not have a body

Body Cookies Headers 10 Test Results

`{}` JSON `>` Preview `<img alt="visualize icon" data-bbox="238 521 251 531"/> Visualize >`

```
1 {
2   "success": true,
3   "message": "Booking cancelled successfully.",
4   "data": {
5     "id": 2,
6     "user_id": 1,
7     "flight_id": 3,
8     "booking_reference": "BV-TUZIP8CZ3",
```

Phase 3: Passenger CRUD

Same steps:

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:controller API/PassengerController

INFO: Controller [C:\wamp64\www\webInternshipProj\app\Http\Controllers\API\PassengerController.php] created successfully.

Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:request Passenger/StorePassengerRequest

INFO: Request [C:\wamp64\www\webInternshipProj\app\Http\Requests\Passenger\StorePassengerRequest.php] created successfully.
```

StorePassengerRequest:

```

public function authorize(): bool
{
    return auth()->check();
}

public function rules(): array
{
    return [
        'first_name' => ['required', 'string', 'max:100'],
        'last_name' => ['required', 'string', 'max:100'],
        'passport_number' => ['nullable', 'string', 'max:20'],
        'nationality' => ['nullable', 'string', 'max:100'],
        'date_of_birth' => ['nullable', 'date', 'before:today'],
    ];
}

public function messages(): array
{
    return [
        'first_name.required' => 'First name is required.',
        'last_name.required' => 'Last name is required.',
        'date_of_birth.before' => 'Date of birth must be in the past.',
    ];
}

```

PassengerController:

1- List all Passengers for a booking:

```

// -----
// GET /api/bookings/{id}/passengers
// List all passengers for a booking
// -----
public function index($bookingId)
{
    $booking = Booking::find($bookingId);

    if (!$booking) {
        return response()->json([
            'success' => false,
            'message' => 'Booking not found.',
        ], 404);
    }

    // Only the booking owner can see passengers
    if ($booking->user_id !== auth()->id()) {
        return response()->json([
            'success' => false,
            'message' => 'You are not authorized to view passengers for this booking.',
        ], 403);
    }

    return response()->json([
        'success' => true,
        'data' => $booking->passengers,
    ], 200);
}

```

2- Add new Passengers:

only owner of the booking can add passengers:

```

// POST /api/bookings/{id}/passengers
// Add a passenger to a booking
// -----
public function store(StorePassengerRequest $request, $bookingId)
{
    $booking = Booking::find($bookingId);

    if (!$booking) {
        return response()->json([
            'success' => false,
            'message' => 'Booking not found.',
        ], 404);
    }

    // Authorization check
    if ($booking->user_id !== auth()->id()) {
        return response()->json([
            'success' => false,
            'message' => 'You are not authorized to add passengers to this booking.',
        ], 403);
    }

    // Cannot add passengers to a cancelled booking
    if ($booking->status === 'cancelled') {
        return response()->json([
            'success' => false,
            'message' => 'Cannot add passengers to a cancelled booking.',
        ], 422);
    }

    // Check passenger count does not exceed seat count
    $currentPassengers = $booking->passengers()->count();
    if ($currentPassengers >= $booking->seat_count) {
        return response()->json([
            'success' => false,
            'message' => 'Cannot add more passengers than booked seats (' . $booking->seat_count . ').',
        ], 422);
    }

    $passenger = Passenger::create([
        'booking_id' => $booking->id,
        'first_name' => $request->first_name,
        'last_name' => $request->last_name,
        'passport_number' => $request->passport_number,
        'nationality' => $request->nationality,
        'date_of_birth' => $request->date_of_birth,
    ]);

    return response()->json([
        'success' => true,
        'message' => 'Passenger added successfully.',
        'data' => $passenger,
    ], 201);
}

```

3- Delete Passenger:

```

// -----
// DELETE /api/bookings/{id}/passengers/{pid}
// Remove a passenger from a booking
// -----
public function destroy($bookingId, $passengerId)
{
    $booking = Booking::find($bookingId);

    if (!$booking) {
        return response()->json([
            'success' => false,
            'message' => 'Booking not found.',
        ], 404);
    }

    if ($booking->user_id !== auth()->id()) {
        return response()->json([
            'success' => false,
            'message' => 'You are not authorized to remove passengers from this booking.',
        ], 403);
    }

    $passenger = Passenger::where('id', $passengerId)
        ->where('booking_id', $bookingId)
        ->first();

    if (!$passenger) {
        return response()->json([
            'success' => false,

```

```

    if (!$passenger) {
        return response()->json([
            'success' => false,
            'message' => 'Passenger not found in this booking.',
        ], 404);
    }

    $passenger->delete();

    return response()->json([
        'success' => true,
        'message' => 'Passenger removed successfully.',
    ], 200);
}

```

The routes:

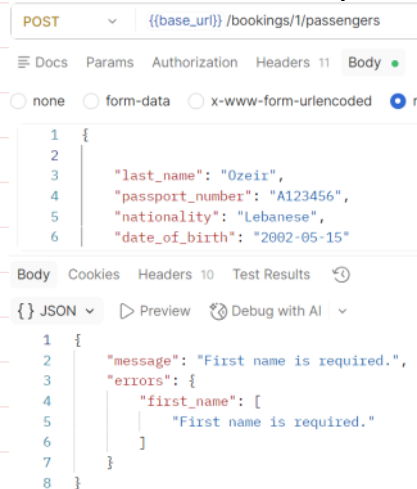
```

// Passengers
Route::get('/bookings/{id}/passengers', [PassengerController::class, 'index']);
Route::post('/bookings/{id}/passengers', [PassengerController::class, 'store']);
Route::delete('/bookings/{id}/passengers/{pid}', [PassengerController::class, 'destroy']);

```

Passenger requests Testing:

Error: First Name required



The booking with id:1 , it contains 3 seat_counts.
I tried to add 4:

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSC

```
1 {
2   "first_name": "Elissar",
3   "last_name": "Ozeir",
4   "passport_number": "A123789",
5   "nationality": "Lebanese",
6   "date_of_birth": "1996-05-13"
7 }
```

Body Cookies Headers 10 Test Results

422 Unprocessable Entity

{ } JSON Preview Debug with AI

```
1 {
2   "success": false,
3   "message": "Cannot add more passengers than booked seats (3).",
4 }
```

Invalid date_of_birth:

```
  "nationality": "Lebanese",
  "date_of_birth": "1996-13-05"
}
```

Cookies Headers 10 Test Results

422 Unprocessable Entity

Preview Debug with AI

```
{
  "message": "The date of birth field must be a valid date.",
  "errors": {
    "date_of_birth": [
      "The date of birth field must be a valid date."
    ]
  }
}
```

Passenger Deleted:

DELETE {{base_url}} /bookings/1/passengers/3

Docs Params Authorization Headers 9 Body Scripts Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ text

1

Body Cookies Headers 10 Test Results

{ } JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Passenger removed successfully."
4 }
```

Adding the saved flights concept (many-to-many relationship):

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:controller API/SavedFlightController

INFO Controller [C:\wamp64\www\webInternshipProj\app\Http\Controllers\API\SavedFlightController.php] created successfully.
```

SavedFlightController:

1- List all saved flights {index}

```
// -----
// GET /api/saved-flights
// List all saved flights for the authenticated user
// -----
public function index(Request $request)
{
    $savedFlights = $request->user()...
    // If the user hasn't saved any flights yet
    if ($savedFlights->isEmpty()) {
        return response()->json([
            'success' => true,
            'message' => "You haven't saved any flight yet.",
            'data' => [],
        ], 200);
    }

    // Normal response with the list
    return response()->json([
        'success' => true,
        'data' => $savedFlights,
    ], 200);
}
```

2- Save a specific flight {save}

```
// -----
// POST /api/flights/{id}/save
// Save a specific flight
// -----
public function save(Request $request, $id)
{
    $flight = Flight::with(['airline', 'originAirport', 'destinationAirport'])
        ->find($id);

    if (!$flight) {
        return response()->json([
            'success' => false,
            'message' => 'Flight not found.',
        ], 404);
    }

    $request->user()->savedFlights()->syncWithoutDetaching([$flight->id]);

    return response()->json([
        'success' => true,
        'message' => 'Flight saved successfully.',
        'data' => $flight,
    ], 200);
}
```

Why syncWithoutDetaching?

It adds the flight if not already saved, but doesn't remove others. Thanks to the unique(user_id, flight_id) constraint, we can never insert duplicates.

3- Unsave a specific flight

```
// -----
// DELETE /api/flights/{id}/unsave
// Delete a specific flight
// -----
public function unsave(Request $request, $id)
{
    $flight = Flight::find($id);

    if (!$flight) {
        return response()->json([
            'success' => false,
            'message' => 'Flight not found.',
        ], 404);
    }

    $request->user()->savedFlights()->detach($flight->id);

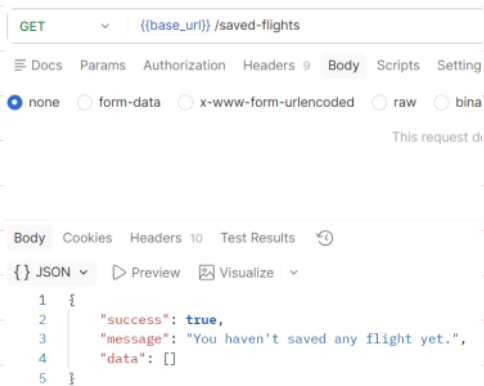
    return response()->json([
        'success' => true,
        'message' => 'Flight removed from saved list.',
    ], 200);
}
```

Routes:

```
// Saved flights
Route::get('/saved-flights', [SavedFlightController::class, 'index']);
Route::post('/flights/{id}/save', [SavedFlightController::class, 'save']);
Route::delete('/flights/{id}/unsave', [SavedFlightController::class, 'unsave']);
```

Testing:

- List Flights:



- Save Flight:

```
public function save(Request $request, $id)
{
    $flight = Flight::with(['airline', 'originAirport', 'destinationAirport'])
        ->find($id);

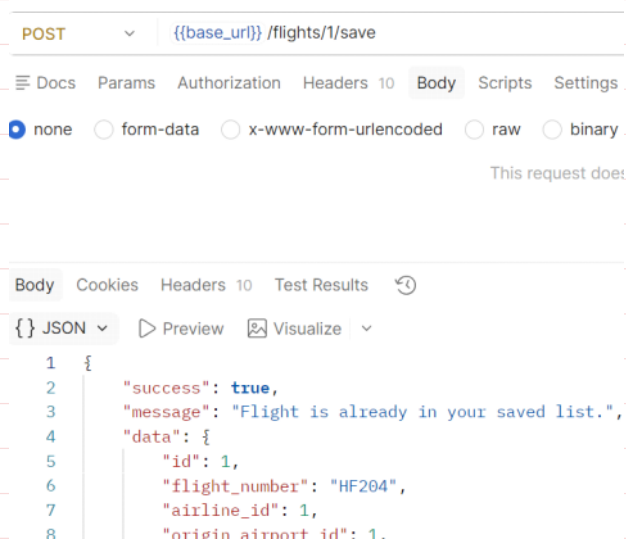
    if (!$flight) {
        return response()->json([
            'success' => false,
            'message' => 'Flight not found.',
        ], 404);
    }

    $user = $request->user();
    $alreadySaved = $user->savdFlights()->where('flight_id', $flight->id)->exists();
    if ($alreadySaved) {
        return response()->json([
            'success' => true,
            'message' => 'Flight is already in your saved list.',
            'data' => $flight,
        ], 200);
    }

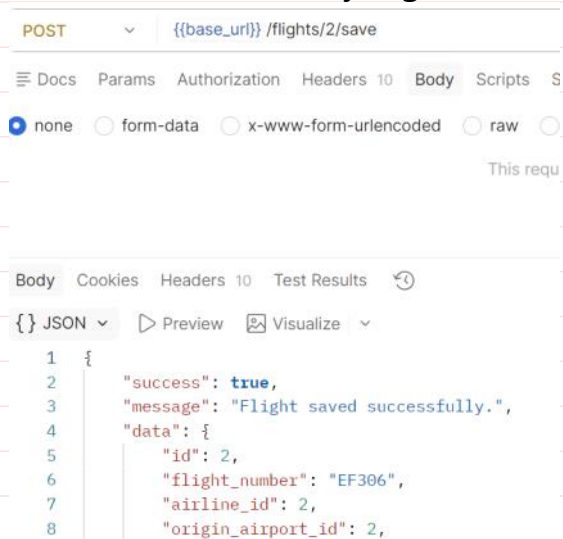
    $user->savdFlights()->attach($flight->id); // Not saved yet → attach it

    return response()->json([
        'success' => true,
        'message' => 'Flight saved successfully.',
        'data' => $flight,
    ], 200);
}
```

When I save the same flight another time, it gives me:

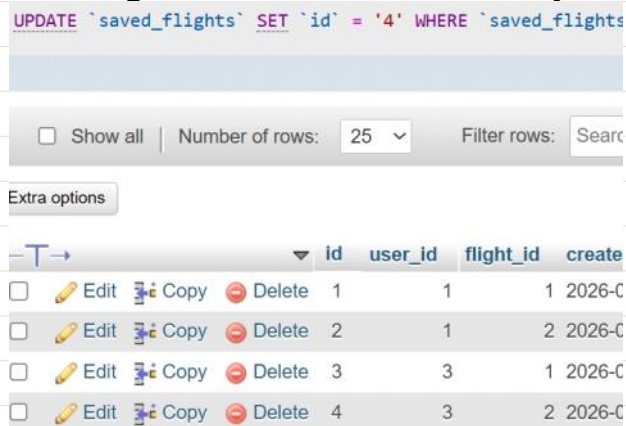


A user can save many flights:



This user has booked 2 flights, with id 1 and 2.

I will login as another user, and try to save these 2 same flights.



This photo of the saved_flights database shows that:

The user with id:1, saved 2 flights (id 1 and 2)

Also the user with id:3, saved the same 2 flights (id 1 and 2)

It shows clearly the many-to-many relationship between the 2 tables: user and flights.

Now I will create the ProfileController and the requests (getProfile, updateProfile)

```
Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:controller API/ProfileController

INFO: Controller [C:\wamp64\www\webInternshipProj\app\Http\Controllers\API\ProfileController.php] created successfully.

Administrator@Mediane MINGW64 /c/wamp64/www/webInternshipProj (develop)
$ php artisan make:request Profile/UpdateProfileRequest

INFO: Request [C:\wamp64\www\webInternshipProj\app\Http\Requests\Profile\UpdateProfileRequest.php] created successfully.
```

UpdateProfileRequest:

```
public function authorize(): bool
{
    return auth()->check();
}

public function rules(): array
{
    return [
        'name' => ['sometimes', 'string', 'min:1', 'max:255'],
        'phone' => ['nullable', 'string', 'max:14'],
        'password' => ['nullable', 'string', 'min:8', 'confirmed'],
    ];
}

public function messages(): array
{
    return [
        'name.required' => 'Your name cannot be empty.',
        'name.max' => 'Name must not exceed 255 characters.',
        'phone.max' => 'Phone number is too long.',
        'password.min' => 'New password must be at least 8 characters.',
        'password.confirmed' => 'Password confirmation does not match.',
    ];
}
```

ProfileController:

1- Get the Profile {show}

```
// -----
// GET /api/profile
// Get profile of the authenticated user
// -----
public function show(Request $request)
{
    return response()->json([
        'success' => true,
        'data' => $request->user(),
    ], 200);
}
```

2- Change the Profile {update}

```
// -----
// PUT /api/profile
// Change the profile
// -----
public function update(UpdateProfileRequest $request)
{
    $user = $request->user();

    // Collect only the fields we allow to update
    $data = $request->only(['name', 'phone']);

    // If a new password is provided, hash it automatically via User model cast
    if ($request->filled('password')) {
        $data['password'] = $request->password;
    }

    $user->update($data);

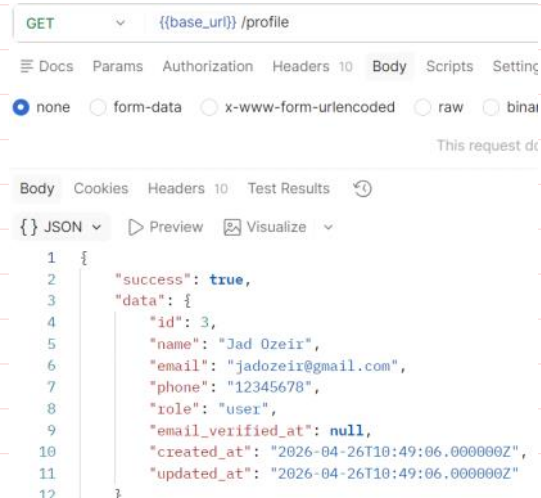
    return response()->json([
        'success' => true,
        'message' => 'Profile updated successfully.',
        'data' => $user,
    ], 200);
}
```

Routes:

```
// Profile
Route::get('/profile', [ProfileController::class, 'show']);
Route::put('/profile', [ProfileController::class, 'update']);
```

Testing:

Get Profile:



The screenshot shows a REST client interface with a GET request to `{{base_url}}/profile`. The response is in JSON format, showing a successful update of a user profile.

```
{
  "success": true,
  "data": {
    "id": 3,
    "name": "Jad Ozeir",
    "email": "jadozeir@gmail.com",
    "phone": "12345678",
    "role": "user",
    "email_verified_at": null,
    "created_at": "2026-04-26T10:49:06.000000Z",
    "updated_at": "2026-04-26T10:49:06.000000Z"
  }
}
```

Another user:

```
GET {{base_url}} /profile

Body
  none
  form-data
  x-www-form-urlencoded
  raw

Body
  Cookies
  Headers 10
  Test Results

JSON
  Preview
  Visualize

1 {
2   "success": true,
3   "data": {
4     "id": 1,
5     "name": "Mediane Ozeir",
6     "email": "mediane@gmail.com",
7     "phone": "12345678",
8     "role": "admin",
9     "email_verified_at": null,
```

Change Profile:

```
PUT {{base_url}} /profile

Body
  none
  form-data
  x-www-form-urlencoded
  raw

1 {
2   "name": "Mediane Ghassan Ozeir"
3 }

Body
  Cookies
  Headers 10
  Test Results

JSON
  Preview
  Visualize

1 {
2   "success": true,
3   "message": "Profile updated successfully.",
4   "data": {
5     "id": 1,
6     "name": "Mediane Ghassan Ozeir",
7     "email": "mediane@gmail.com",
8     "phone": "12345678",
```

I haven't change the phone, it remains the same

Changing only the phone number:

```
PUT {{base_url}} /profile

Body
  none
  form-data
  x-www-form-urlencoded
  raw

1 {
2   "phone": "12345678"
3 }

Body
  Cookies
  Headers 10
  Test Results

JSON
  Preview
  Visualize

1 {
2   "success": true,
3   "message": "Profile updated successfully.",
4   "data": {
5     "id": 1,
6     "name": "Mediane Ghassan Ozeir",
```

We can't leave the name empty:

```
{
  "name": "",
  "phone": "12345678"
}
```

Cookies Headers 10 Test Results 

422 Unprocessable

DN   Preview  Debug with AI 

```
{
  "message": "The name field must be a string. (and 1 more error)",
  "errors": {
    "name": [
      "The name field must be a string.",
      "The name field must be at least 1 characters."
    ]
  }
}
```

Ba3ed badde zid:

- Bas khallis badde aamil Admin Panel